CS402 Theory of Automata

Question No : 19 of 30

In PDA, an alphabet of STACK characters is represented by the symbol _____

Answer (Please select your correct option)

- Σ
- Γ
- $
- Δ

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S402 Theory of Automata

Question No : 21 of 38

A language can be represented by _____ RE / RE(s).

Answer (Please select your correct option)

- more than one
- only one
- only two
- only three

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Question No : 22 of 38

A is not a valid transition in _____

Answer (Please select your correct option)

- TG
- GTG
- NFA
- ɜE

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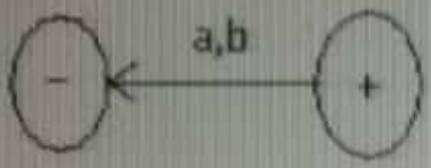
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CS402 Theory of Automata

Question No : 20 of 38

Which language will be accepted by the given Finite Automaton (FA) ?



Answer (Please select your correct option)

- Having equal number of a's and b's
- It does not accept any language
- Having double a's followed by double b's
- Having one 'a' and bbb

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CS402 Theory of Automata

Question No : 24 of 30

A nonregular language, by _____ theorem, can't be accepted by any FA or TG.

Answer (Please select your correct option)

- Palindrome
- Associative
- Closure
- Kleene's

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Question No : 25 of 38

What will be the Regular Expression for the following CFG?

S→aX| bX

X→a

X→b

X→^

Answer (Please select your correct option)

- ab
- (a+b)*
- a*b*
- (a+b)+

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Question No : 27 of 38

In context of new format of "FA" _____ state does not has outgoing edge(s)

Answer (Please select your correct option)

- Start
- Read
- Pop
- Accept

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CS402 Theory of Automata

Question No : 30 of 38

Grammatical rules which are concerned with the structure is called _____.

Answer (Please select your correct option)

- Semantic
- Non-terminal
- Syntax
- Terminal

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CS402 Theory of Automata

Question No : 29 of 38

By applying method II of Equivalent languages, an FA accepts no string if _____

Answer (Please select your correct option)

- Final state is not marked
- Initial state is not marked
- Final state is marked
- None of the given

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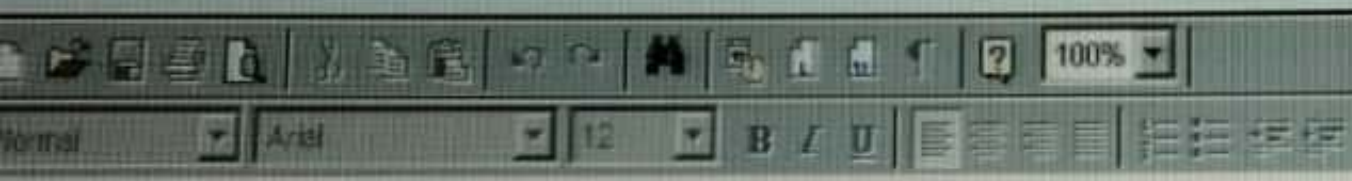
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CS402 Theory of Automata

Question No : 32 of 39

What is meant by "productions" in the context of Context Free Grammar (CFG)? Give an example.

Answer (Please click here to Add Answer)



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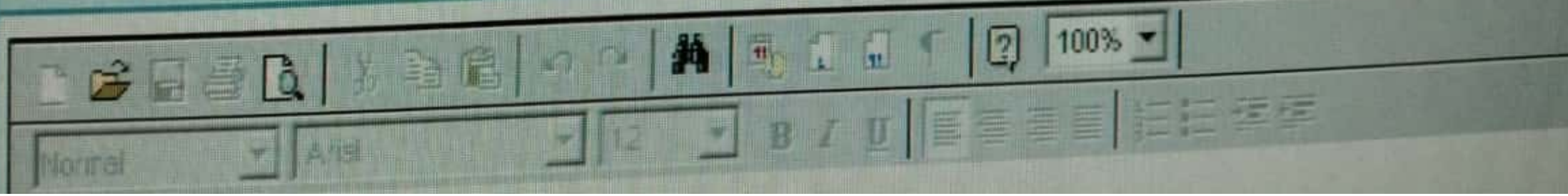
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Question No : 35 of 38

Find Pref(Q in R) where Q = { bbbb,bbbaa,bba} and R = {aa,abaaabb,bbbbbaa,bbbbbbbbbb}

Answer (Please click here to Add Answer)



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CS402 Theory of Automata

Question No : 34 of 38

Identify the words which are accepted by the following CFG.

S---->abA
A---->baB
B---->aA|bb

1. abbaababba
2. abbaabb
3. abbbaabb
4. abbaababb
5. abbaabaababb

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CS402 Theory of Automata

Question No : 36 of 38

Develop corresponding Context Free Grammar (CFG) for the regular expression given below:

$a(a+b)^*b$

Answer (Please click here to Add Answer)

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CS402 Theory of Automata

Question No : 37 of 38

Draw an NFA which represents the following language.

L = { bab, abab, bbab, aabab, bbbab, abbab, babab .....

Answer (Please click here to Add Answer)

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Kleene star closure is a _____ operation.

Answer (Please select your correct option)

- unary
- binary
- tertiary
- multiplicative

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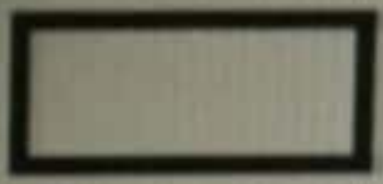
Question No : 3 of 30

If a language L accepts all words of odd length then L^c will accept _____.

Answer (Please select your correct option)

- Infinite number of words
- All words of prime length
- All words of odd length
- All words of even length

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Above symbol is used for _____ state in PDA.

Answer (Please select your correct option)

- Pop
- Reject
- Push
- Read

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CS402 Theory of Automata

Question No : 7 of 38

Before the CFG corresponding to any PDA is determined, the PDA should be converted into _____

Answer (Please select your correct option)

- Conversion Form
- Turing Machine
- Regular Expression
- Chomsky Normal Form

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Question No : 9 of 30

Pumping lemma divides a string into _____ parts.

Answer (Please select your correct option)

- one
- two
- three
- four

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402 Theory of Automata

estion No : 11 of 38 Mark

he CFG is said to be ambiguous if there exists at least one word of its language that can be generated by _____ production tree(s)

nswer (Please select your correct option)

- more than one
- at least one
- one
- at most one

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CS402 Theory of Automata

Question No : 15 of 38

The process of finding the derivation of word generated by particular grammar is called _____

Answer (Please select your correct option)

- Processing
- Parsing
- Programming
- Planning

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Question No : 13 of 38

The CFG **S –> aSa | bSb | a | b | ^** represents the language _____.

Answer (Please select your correct option)

- PALINDROME
- EQUAL
- ODD-ODD
- EVEN-EVEN

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CS402 Theory of Automata

Question No : 16 of 38

The language of Palindromes defined over an alphabet set {a,b} can be recognized by _____

Answer (Please select your correct option)

- FA
- NFA
- TG
- PDA

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Question No : 18 of 38

The structure given below is called _____.

S -> aA|bB
A -> aS|a
B -> bS|b

Answer (Please select your correct option)

- RE
- CFG
- FDA
- TG

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Question No : 14 of 38

If an FA has N states then it must accept the word of length _____

Answer (Please select your correct option)

- N-1
- N+1
- N
- 2N

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S402 Theory of Automata

Question No : 17 of 38

Let A = (0, 1). The number of possible strings of length 'n' that can be formed by the elements of the set A is ____

Answer (Please select your correct option)

- n!
- n^2
- n^n
- 2^n

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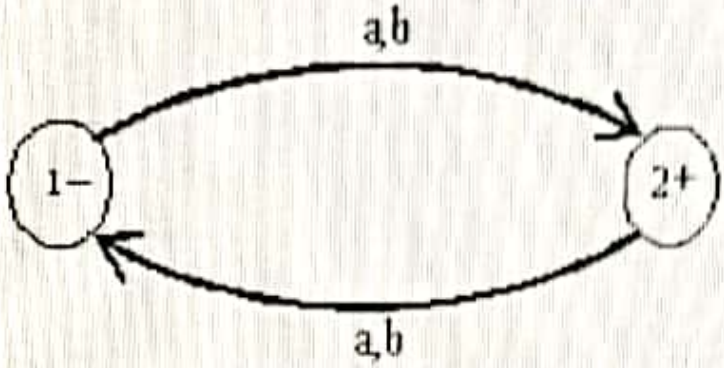
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Question No : 12 of 38

Below given FA accepts the string _____.



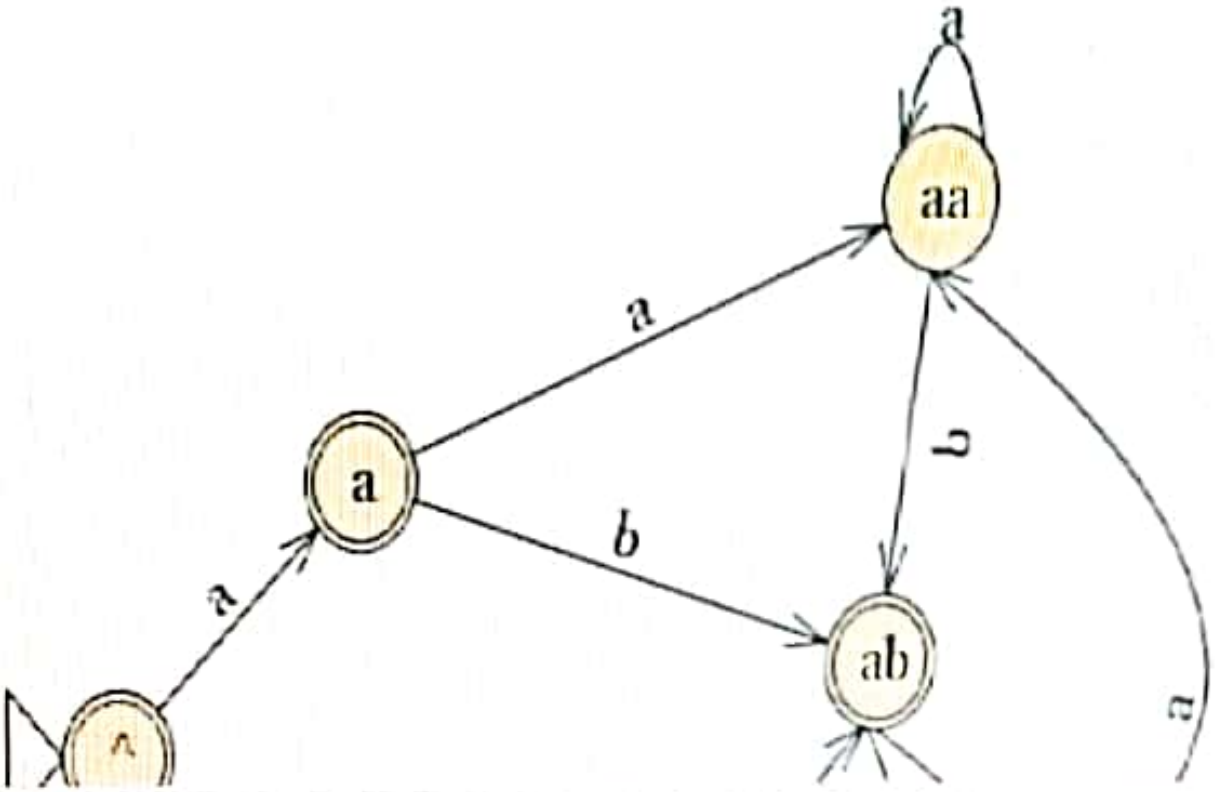
Answer (Please select your correct option)

- abbb
- abab
- aaab
- aaa

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Question No : 10 of 38

Below FA is drawn using _____ structure.



Answer (Please select your correct option)

- graph
- tree
- simple
- complex

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CS402 Theory of Automata

Question No : 8 of 30

In Row Language, CFG is introduced in the form Net (A, B, C), where A and B are _____ and C is _____

Answer (Please select your correct option)

- stack characters, joint
- pop characters, joint
- joints, stack character
- joints, push character

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CS402 Theory of Automata

Question No : 5 of 30

Left hand side of a production in Context Free Grammer (CFG) consists of _____

Answer (Please select your correct option)

- more than one terminal
- one non-terminal
- terminals and non-terminals
- one terminal

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CS402 Theory of Automata

Question No : 4 of 30

Can a turing machine's head ever be in the same location in two successive steps?

Answer (Please select your correct option)

- Yes
- Yes but only in finite languages
- No
- Yes but only in infinite langauges

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CS402 Theory of Automata

There can be more than _____ FA for a certain language but for _____ FA there is only one language associated with it.

Answer (Please select your correct option)

- one, one
- one, two
- two, three
- two, one

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CS402 Theory of Automata

Question No : 33 of 38 **Marks: 3 (Budget**

Consider the language L of string, defined over $\sum = \{0,1\}$, having double 0's or double 1's. What are the strings that cannot be accepted by this language?

Answer (Please click here to Add Answer)



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5402 Theory of Automata

Question No : 38 of 38

Explain any two rules to define all possible productions of Context Free Grammar (CFG) of "Row language".

Answer (Please click here to Add Answer)

100%

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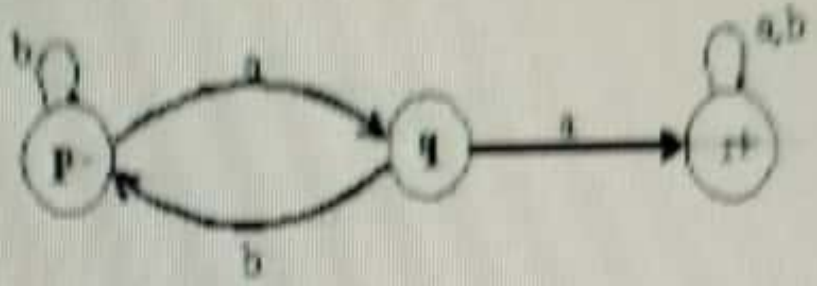
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CS402 Theory of Automata

Question No : 31 of 38

Draw the complement for the following FA



Answer (Please click here to Add Answer)

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CS402 Theory of Automata

Question No : 28 of 30

The language Q is said to be quotient of two regular languages P and R, denoted by_____

Answer (Please select your correct option)

- Q=R/P if P/Q=R
- Q=R/P if PQ=R
- Q=R*P if PQ=R
- Q=R+P if P*Q=R

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CS402 Theory of Automata

Question No : 25 of 30

What will be the Regular Expression for the following CFG?

S → aX | bX

X → a

X → b

X → ε

Answers (Please select your correct option)

- a
- (a+b)*
- a*b*
- (a+b)a

CS402 Theory of Automata

Question No : 26 of 38

A production of the form nonterminal ⟶ terminal is called a _____.

Answer (Please select your correct option)

- Dead Production
- Live Production
- Empty Production
- Waste Production

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CS402 Theory of Automata

Question No : 23 of 38

We can have more than one parse tree for a given string in case of _____ Context Free Grammar (CFG)

Answer (Please select your correct option)

- complicated
- ambiguous
- lazy
- simple

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